



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

LAB ASSIGNMENT-1

Submitted by:

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1. Write a program (WAP) to display "Hello World" on console display.

```
new.cpp x
OOPS > new.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      cout<<"Hello world";
6  }
```

OUTPUT:

```
PROBLEMS OUTPUT DEBUG CONSOLE
PS C:\codes\UNC401> cd "c:\code
● Hello world
○ PS C:\codes\UNC401\OOPS>
```

2. WAP to that reads two values from keyboard and displays the larger value on the screen.

```
OOPS > new.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int a,b;//a=5,b=3
6      cin>>a>>b;
7      int x=a>b?a:b;
8
9      cout<<x<<endl;
10 }
```

OUTPUT:

```
PS C:\codes\UNC401\OOPS> cd "c
cd "c
5 4
5
PS C:\codes\UNC401\OOPS>
```

3. WAP to read three values a, b, c from keyboard and display the value of $x = a/b - c$.

```
new.cpp
OOPS > new.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int a,b,c;
6      cin>>a>>b>>c;
7      double x=a/b-c;
8      cout<<x<<endl;
9  }
```

OUTPUT:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PS C:\codes\UNC401> cd "c:\codes\UNC401\O
● 10
  4
  5
  -3
○ PS C:\codes\UNC401\OOPS>
```

4.WAP to implement the following control characters:

- '\n' is for new line, or you can use endl – cout<<endl<<"message";
- '\t' is for tab
- '\a' is an alarm sound
- '\r' is carriage return to go to the beginning of the current line

```
OOPS > new.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      cout<<"My name is ryan. \n I live in amritsar"<<endl;
6      cout<<"I am \t 19 years old"<<endl;
7      cout<<"\a"<<endl;
8      cout<<"\r end"<<endl;
9  }
```

OUTPUT:

```
cd "c
My name is ryan.
I live in amritsar
I am      19 years old

end
```

5. Implement *namespace* in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.

```
1  #include<iostream>
2  using namespace std;
3
4
5  namespace first{
6      int x=1;
7      void display(){
8          cout<<"The value of x is :"<<x<<endl;
9      }
10 }
11 namespace second{
12     int x=2;
13     void display(){
14         cout<<"The value of x is :"<<x<<endl;
15     }
16 }
17
18 int main(){
19     cout<<first::x<<endl;
20     cout<<second::x<<endl;
21     first::display();
22     second::display();
23 }
```

OUTPUT:

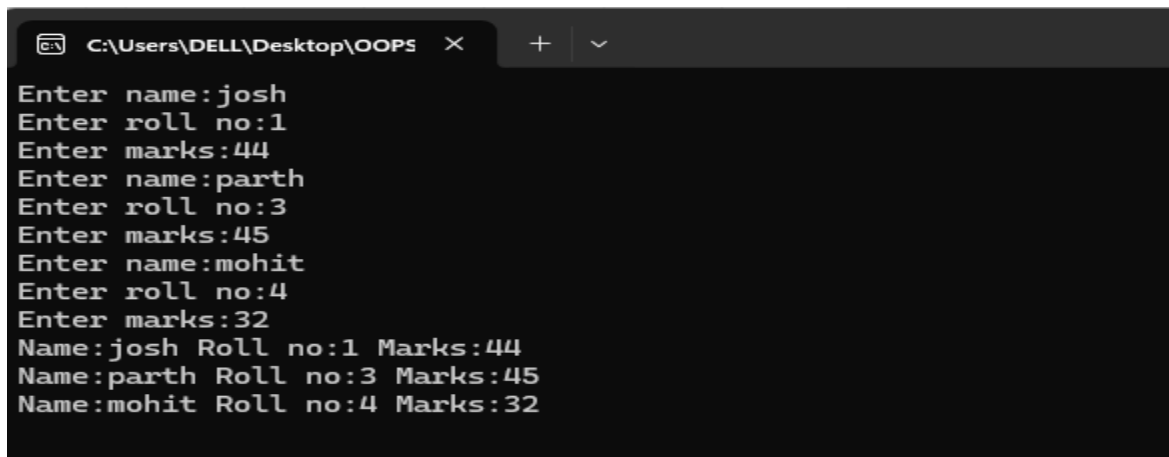
```
PS C:\codes\UNC401> cd "c:\codes
1
2
○ The value of x is :1
The value of x is :2
PS C:\codes\UNC401\OOPS>
```

6. Write a program to define a structure *Student* that contains 3 variables and 2 functions:

- Variables: (a) name (b) roll (c) marks
- Functions: (1) *setStudentData()* (2) *getStudentData()*

```
OOPS > new.cpp > ...
1  #include<iostream>
2  using namespace std;
3
4  struct Student{
5      char name[15];
6      int roll;
7      float marks;
8
9      void setStudentData(){
10         cout<<"Enter name:";
11         cin>>name;
12         cout<<"Enter roll no:";
13         cin>>roll;
14         cout<<"Enter marks:";
15         cin>>marks;
16     }
17
18     void getStudentData(){
19         cout<<"Name:"<<name;
20         cout<<" Roll no:"<<roll;
21         cout<<" Marks:"<<marks<<endl;
22     }
23 };
24
25 int main(){
26     Student s[3];
27
28     for(int i=0;i<3;i++)
29         s[i].setStudentData();
30
31     for(int i=0;i<3;i++)
32         s[i].getStudentData();
33
34     return 0;
35 }
36
```

OUTPUT:



```
C:\Users\DELL\Desktop\OOPS > Enter name:josh
Enter roll no:1
Enter marks:44
Enter name:parth
Enter roll no:3
Enter marks:45
Enter name:mohit
Enter roll no:4
Enter marks:32
Name:josh Roll no:1 Marks:44
Name:parth Roll no:3 Marks:45
Name:mohit Roll no:4 Marks:32
```

7. In the previous program make the data private, and functions public.

```
OOPS > G+ new.cpp > ...
1  #include<iostream>
2  using namespace std;
3
4  class Student{
5  private:
6      char name[15];
7      int roll;
8      float marks;
9
10 public:
11     void setStudentData(){
12         cout<<"Enter name:";
13         cin>>name;
14         cout<<"Enter roll no:";
15         cin>>roll;
16         cout<<"Enter marks:";
17         cin>>marks;
18     }
19
20     void getStudentData(){
21         cout<<"Name:"<<name;
22         cout<<" Roll no:"<<roll;
23         cout<<" Marks:"<<marks<<endl;
24     }
25 };
26
27 int main(){
28     Student s[3];
29
30     for(int i=0;i<3;i++)
31         s[i].setStudentData();
32
33     for(int i=0;i<3;i++)
34         s[i].getStudentData();
35
36     return 0;
37 }
38
```

OUTPUT:



C:\Users\DELL\Desktop\OOPS



Enter name:rohan

Enter roll no:1

Enter marks:22

Enter name:Mohit

Enter roll no:2

Enter marks:56

Enter name:Farak

Enter roll no:3

Enter marks:89

Name:rohan Roll no:1 Marks:22

Name:Mohit Roll no:2 Marks:56

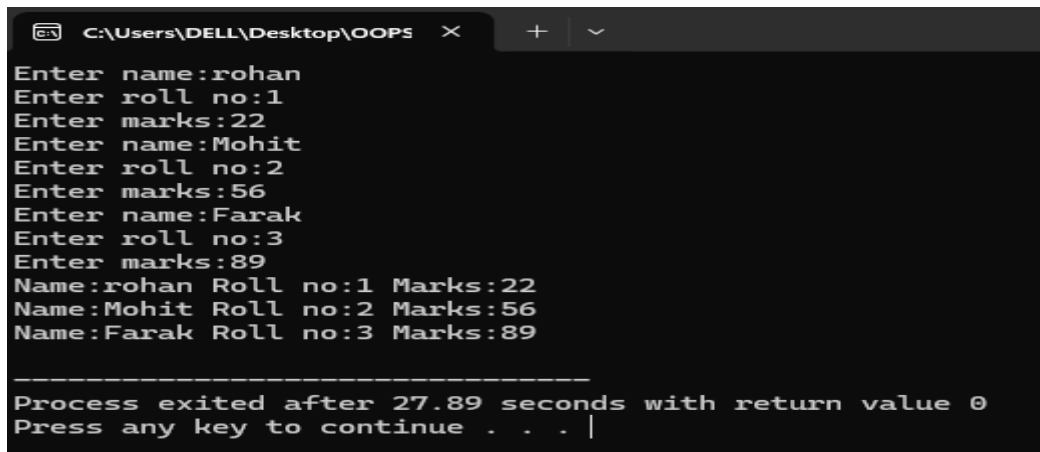
Name:Farak Roll no:3 Marks:89

Process exited after 27.89 seconds with return value 0
Press any key to continue . . . |

8. Convert program #4 from struct to class.

```
OOPS > G new.cpp > ...
1  #include<iostream>
2  using namespace std;
3
4  class Student{
5  private:
6      char name[15];
7      int roll;
8      float marks;
9
10 public:
11     void setStudentData(){
12         cout<<"Enter name:";
13         cin>>name;
14         cout<<"Enter roll no:";
15         cin>>roll;
16         cout<<"Enter marks:";
17         cin>>marks;
18     }
19
20     void getStudentData(){
21         cout<<"Name:"<<name;
22         cout<<" Roll no:"<<roll;
23         cout<<" Marks:"<<marks<<endl;
24     }
25 };
26
27 int main(){
28     Student s[3];
29
30     for(int i=0;i<3;i++)
31         s[i].setStudentData();
32
33     for(int i=0;i<3;i++)
34         s[i].getStudentData();
35
36     return 0;
37 }
38
```

OUTPUT:



```
C:\Users\DELL\Desktop\OOPS >
Enter name:rohan
Enter roll no:1
Enter marks:22
Enter name:Mohit
Enter roll no:2
Enter marks:56
Enter name:Farak
Enter roll no:3
Enter marks:89
Name:rohan Roll no:1 Marks:22
Name:Mohit Roll no:2 Marks:56
Name:Farak Roll no:3 Marks:89
-----
Process exited after 27.89 seconds with return value 0
Press any key to continue . . . |
```

9. Discuss the differences of structures in C and structures in C++ during the lab with other students or your instructor.

Structure in C

1. Structures in C are used only to store data; they cannot have functions inside them.
2. C structures do not have access specifiers, so all data members are openly accessible.
3. Since there is no access control, data hiding is not possible in C structures.
4. C structures do not support constructors or destructors for automatic initialization and cleanup.
5. Overall, C structures do not support object-oriented programming concepts.

Structure in C++

1. In C++, structures can contain both data members and functions.
2. C++ structures allow the use of access specifiers like public and private to control access.
3. Because of access control, data hiding is supported in C++ structures.

4. C++ structures support constructors and destructors, making object initialization easier.
5. C++ structures support object-oriented programming concepts such as encapsulation and inheritance.

10. WAP to read an integer value from the keyboard and display on screen "WELL DONE" that many times.

```
OOPS > G+ new.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cout << "How many times you want to print well done?\n";
7      cin >> n;
8
9      for (int i = 1; i <= n; i++)
10         cout << "WELL DONE" << endl;
11     return 0;
12 }
13
```

OUTPUT:

```
● PS C:\codes\UNC401\OOPS> cd "c:\codes\UNC401\OOPS\"
How many times you want to print well done?
5
WELL DONE
WELL DONE
WELL DONE
WELL DONE
WELL DONE
```

11. WAP to utilize assignment += and -= operators.

```
new.cpp
OOPS > new.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int x, y;
6      cout << "Enter value for x:";
7      cin >> x;
8      cout << "Enter value for y:";
9      cin >> y;
10     x += y;
11     cout << "After x+=y,x=" << x << endl;
12
13     x -= y;
14     cout << "After x-=y,x=" << x << endl;
15
16     return 0;
17 }
18
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS x + v
Enter value for x:33
Enter value for y:55
After x+=y,x=88
After x-=y,x=33
-----
Process exited after 1.739 seconds with return value 0
Press any key to continue . . . |
```

12.Swap two numbers using bitwise operator.

```
OOPS > G+ new.cpp > ...
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int a,b;
6      cout<<"Enter value for a:";
7      cin>>a;
8      cout<<"Enter value for b:";
9      cin>>b;
10
11     a=a^b;
12     b=a^b;
13     a=a^b;
14
15     cout<<"After swapping:"<<endl;
16     cout<<"a="<<a<<endl;
17     cout<<"b="<<b<<endl;
18     return 0;
19 }
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS X + v
Enter value for a:45
Enter value for b:78
After swapping:
a=78
b=45

-----
Process exited after 2.98 seconds with return value 0
Press any key to continue . . . |
```

13. Write a program to solve the following problem:

A library charges a fine for every book return late. For first 5 days the fine is 50 paise, for 6–10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled. WAP to accept number of days the member is late to return the book and display the fine or appropriate message.

```
OOPS > G+ new.cpp > ...
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int days;
6      cout<<"How many days was he delay?\n";
7      cin>>days;
8
9      if(days<=5){
10         cout<<"Fine=0.50 rupees";
11     }
12     else if(days<=10){
13         cout<<"Fine=1.00 rupees";
14     }
15     else if(days<=30){
16         cout<<"Fine=5.00rupees";
17     }
18     else{
19         cout<<"Membership cancelled";
20     }
21     return 0;
22 }
23
```

OUTPUT:



C:\Users\DELL\Desktop\OOPS



How many days was he delay?

34

Membership cancelled

Process exited after 2.158 seconds with return value 0

Press any key to continue . . . |